ROLLUPOS APPLICATION USER MANUAL • AUTOMOTIVE & TELEMETRY

⚡ Lightning Garage

2023 F-150 Lightning ER Lariat FORScan Mod Menu, Deterministic As-Built .abt/.ab Parser & Checksum Companion

`Version: v1.0.0` `Build Hash: #6fed71f7` `🔒 Per-User Authorized Cartridge` `Auto-Synced: 2026-10-26 16:43`

1. ⚡ 60-Second Quickstart Guide

1 **Verify Your Truck & VIN (`⚡ Truck & VIN`)**
Launch Lightning Garage to view your pre-loaded 2023 F-150 Lightning ER Lariat (`1FT6W1EV4PWG11434`, 131 kWh Extended Range, 320 mi EPA). Tap `🌐 Verify w/ NHTSA vPIC` anytime to confirm factory BEV telemetry.

2 **Load a Baseline (`.abt` / `.ab` or `💉 Demo Baseline`)**
Tap `📂 Import .abt / .ab File` to load your real FORScan `.abt` module backups or Ford Motorcraft `.ab` XML—or tap `💉 Load Demo Baseline` to preview live nibble mask and checksum dry-runs immediately.

3 **Browse & Star Mods (`🎛️ Mod Menu`)**
Filter all 59 F-150 recipes by `🔥 Most Common (★★+)`, module (`726 BCM`, `7D0 APIM`, `720 IPC`, `740/741 DDM/PDM`), or Risk Tier (`T1 Cosmetic`, `T2 Behavior`, `T3 Locked`). Tap any card to inspect exact `BEFORE -> AFTER` hex bytes and recomputed checksums.

4 **Run Your In-Cab FORScan Session (`📋 Session Plan`)**
Open the `📋 Plan` tab in the truck cab with your OBDLink EX. Follow the 12V pre-flight checklist, verify every target module has a saved `.abt` backup, and copy or share your exact line-by-line write sheet.

2. 🧭 Screen-by-Screen & Navigation Map

SCREEN / TAB / MODAL	WHAT IT CONTROLS & HOW TO USE IT
⚡ **Truck & VIN**	Offline ISO 3779 + NHTSA vPIC VIN decoder, readiness KPIs, 1-tap Demo Baseline loader, and JSON Vault export/import.
🎛️ **Mod Menu (59 Recipes)**	Searchable FORScan mod catalog sorted by popularity and risk tier with live nibble-mask & checksum dry-run drawer.
🧩 **Modules (87 Addresses)**	Complete Ford control module directory (`726`, `7D0`, `720`, `706`, `7E7`, `746`, etc.) with EV notes, write-risk ratings, and backup evidence.
🛡️ **Backups & Factory .ab**	Pre-flight module backup coverage checker, `.abt`/`.ab` importer, Factory vs. Current byte drift differ, Motorcraft AsBuilt helper, and Checksum Calculator.
📋 **Session Plan**	Printable/shareable in-cab FORScan write sheet grouping required module backups and exact target hex lines.

3. 🔬 In-Depth Feature Guide & Workflows

100% Deterministic Nibble Mask & Checksum Engine (`Code -> Never LLM`)

Evaluates `set`, `bits`, and `offset` nibble operations across multi-step recipes with rolling same-address state compounding and recomputes exact Ford line checksums (`sum(header + data[:-1]) & 0xFF`).

3-Tier Safety Gating & EV Applicability Filtering

Automatically locks Tier 3 safety/HV/powertrain/ADAS modules (`7E7 BECM`, `746 SOBDMC`, `760 ABS`, `730 PSCM`, `706 IPMA`) and flags `ICE_ONLY` gas-engine recipes so they cannot be accidentally scheduled on your Lightning.

Ford Motorcraft `.ab` XML & FORScan `.abt` Diffing

Parses both plain-text FORScan `.abt` backups and official Ford Motorcraft `.ab` XML files directly on your phone, surfacing byte-by-byte factory drift and hardware part numbers (`F111`, `F113`, `F188`).

4. ⚙️ Settings, Secret Vault & Multi-User Data Isolation

SETTING KEY	FIELD DESCRIPTION	TYPE	DEFAULT
`default_vin`	Default 17-Character Truck VIN	`text`	`1FT6W1EV4PWG11434`
`default_trim`	Default Trim Level	`text`	`Lariat`
`default_battery`	Default HV Battery Pack	`text`	`Extended Range (131 kWh)`

🔒 **Storage Namespace Pattern:** `rollup::v1::u::<google-email>::app::lightning-garage::*`

📦 **Active App Storage Keys:** `__user_config__` (Managed automatically)

🔐 **Private On-Device Vault Secrets:** None required (uses Hub session)

Isolated Per-User Storage (RollupBridge) | Android Share Sheet & CSV/JSON Export

5. 🎛️ Live UI Controls Reference & Recent Updates

Interactive Controls Detected in Current Build (`v1.0.0`):

📖 Guide | ⚡ Truck & VIN | 🎛️ Mod Menu (59) | 🧩 Modules (87) | 🛡️ Backups & Factory .ab | 📋 Session Plan (0)

🎛️ Mod Menu | 🧩 Modules | 🛡️ Backups/.ab | 📋 Plan (0)

Pro Tips & Best Practices:

- Always save an untouched FORScan `.abt` backup of every target module before writing changes, and import the post-write `.abt` afterward to confirm exact byte matches.
- Tap `📤 Export Vault (.json)` on the Truck tab to back up your starred mods, personal notes, and imported `.abt`/`.ab` baselines across devices.

Recent Cartridge Update Log (Auto-Tracked on Publish):

- Initial stamped cartridge release in RollupOS Hub.